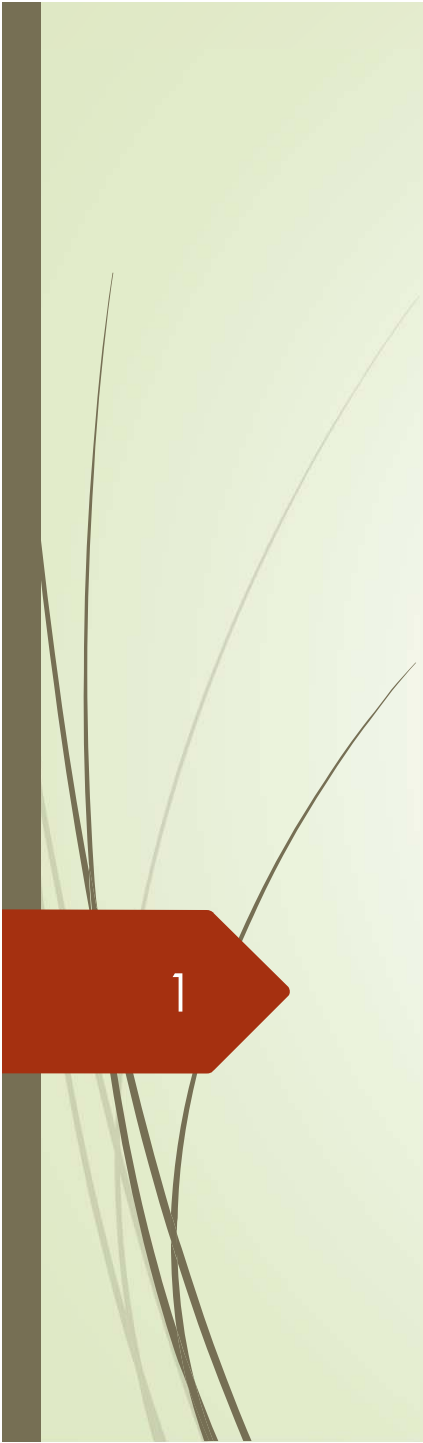




Introduction to **VLSI Design**

1

Lecture# 01

VLSI Design

Course Introduction

- Course Title: VLSI Design
- Course Code:EEE434
- Class: EEE-6
- Text Book
 - Introduction to VLSI Circuits and Systems
 - By John P. Uyemura
- Reference Books
 - VLSI Design
 - Debaprasad Das
 - CMOS VLSI Design
 - Neil H. E. Weste and David Money Harris
- Course Assessment
 - Quizzes (Announced)
 - Assignments (A lot)
 - Exams

Instructor and Content

- Course / Lab Instructor: Dr. Sohaib Ayyaz Qazi
- Contact me at
 - SOHAIB_AYYAZ_QAZI@comsats.edu.pk (please be careful with spellings)
 - Course Material / Announcements will be available at <http://saqazi.com/vlsidesign>
 - Password to access **vlsi434eee6sp19**
 - Office 213, Faculty Block 1, CU Islamabad

Course Learning Outcomes (CLOs)

- Understand and explain the importance of VLSI, design hierarchy, design metrics, IC fabrication materials and IC fabrication processes. **(C2-PLO1)**
- Implement and analyze complex logic circuits and layout (physical design) using static CMOS logic and transmission gates. **(C4-PLO2)**
- Design electrically and geometrically individual MOSFETs, interconnects and other layers as well as logic gates, modules and complete circuits in a CMOS IC. **(C4-PLO3)**
- Apply advanced techniques in CMOS logic circuits (dynamic, dual-rail and domino etc.) and explain reliability of ICs. **(C3-PLO1)**
- Use CAD tools to implement and verify behavioral, RTL, circuit and layout level designs. **(A3-PLO5)**
- Present and analyze data with effective report writing skills. **(A2-PLO10)**

Course Contents

Week #	Lecture Topics
Week 01	Introduction and design metrics
Week 02	Logic design using MOSFETs
Week 03	Physical structure of CMOS ICs
Week 04	CMOS layers, designing MOSFET arrays (logic gates) in layout
Week 05	Fabrication process of CMOS ICs, lithography and design rules
Week 06	Elements of physical design
Week 07	Electrical characteristics of MOSFETs
Week 08	Electronic analysis of CMOS Logic Gates
Week 09	Advances techniques CMOS logic circuits
Week 10	System specifications using VHDL or Verilog
Week 11	General VLSI system components (multiplexors, latches and registers etc.)
Week 12	Arithmetic circuits in CMOS VLSI (Adders and multipliers etc.)
Week 13	Memories architectures and core design
Week 14	System level physical design (delays, crosstalk, scaling & power consumption)
Week 15	Reliability, testing and Design for Test (DFT)
Week 16	Revision

Introduction to VLSI (1/2)

- ▶ VLSI is an acronym for Very Large Scale Integration
 - ▶ Very dense electronic Integrated Circuits (ICs)
 - ▶ Large number of switching devices (transistor) per unit area
- ▶ The number of transistors has exceeded 1 billion per chip (IC)¹
 - ▶ Nowadays up to tens of billions²

Year	Era	Level of Integration
1958	Single Transistor	-
1960	Monolithic IC	1
1962	Multi-function	2-4
1964	Complex function	5-20
1967	Medium scale integration (MSI)	20-200
1972	Large scale integration (LSI)	200-2000
1978	Very large scale integration (VLSI)	2000-20,000
1989	Ultra large scale integration (ULSI)	Above 20,000

¹Peter Clarke, EE Times, "Intel enters billion-transistor processor era", 14 October 2005

²Antone Gonsalves, EE Times, "Samsung begins production of 16-Gb flash", 30 April 2007

Introduction to VLSI (2/2)

- ▶ In order to achieve higher integration density
 - ▶ Transistor sizes are being decreased (scaling)
 - ▶ Transistor scaling is usually referred by channel length
- ▶ Transistor scaling has also raised some issues along with the benefits
 - ▶ Power density
 - ▶ Velocity saturation
 - ▶ Other short channel effects

Parameter	Trends
Number of transistors per IC	Increasing
Transistor size	Decreasing
Operating frequency (speed)	Increasing
Operating voltage	Decreasing (to decrease power consumption)

Transistor scaling and operating frequency reaching its limits. New solutions are being adopted.

Moore's Law

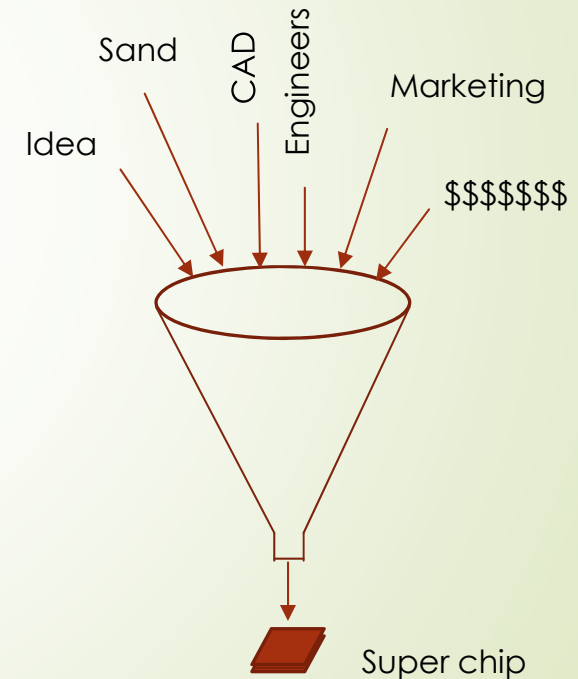
- ▶ Gordon Moore suggested that the number of transistors in an IC
 - ▶ Will double every 18 to 24 months
- ▶ So far, the Moore's law is holding, some what
 - ▶ But it is also struggling
 - ▶ As Gordon Moore himself said in 2015 ¹
 - ▶ *"I guess I see Moore's law dying here in the next decade or so, but that's not surprising."*

¹Moore, Gordon. "Gordon Moore: The Man Whose Name Means Progress, The visionary engineer reflects on 50 years of Moore's Law", March 30, 2015

²Image courtesy <http://www.cringely.com/2013/10/15/breaking-moores-law/> 2016-02-09 at 1100 hrs

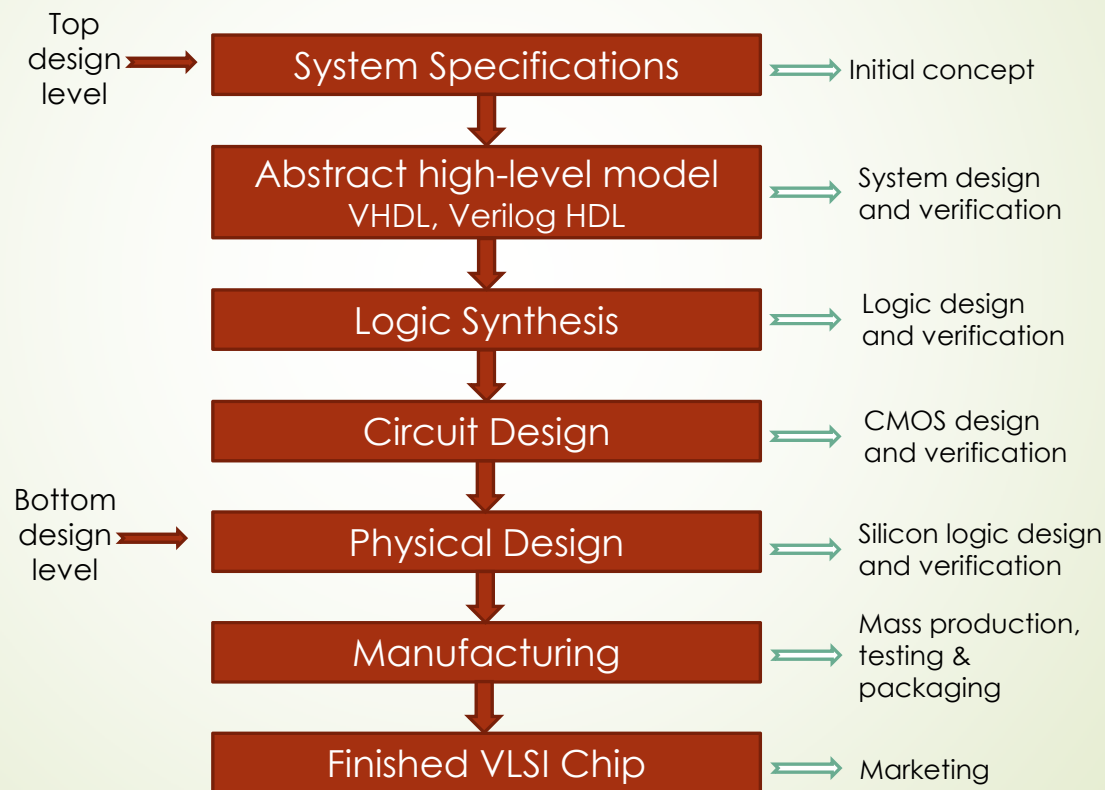
VLSI Complexity and Design (1/2)

- As the integration density increases
 - It also increases design complexity
- So, it is required to have
 - Design team
 - Design hierarchy



VLSI Complexity and Design (2/2)

- Mostly top-down approach
- Bottom-up is only feasible for small projects



Comparison of Approaches

Top - Down

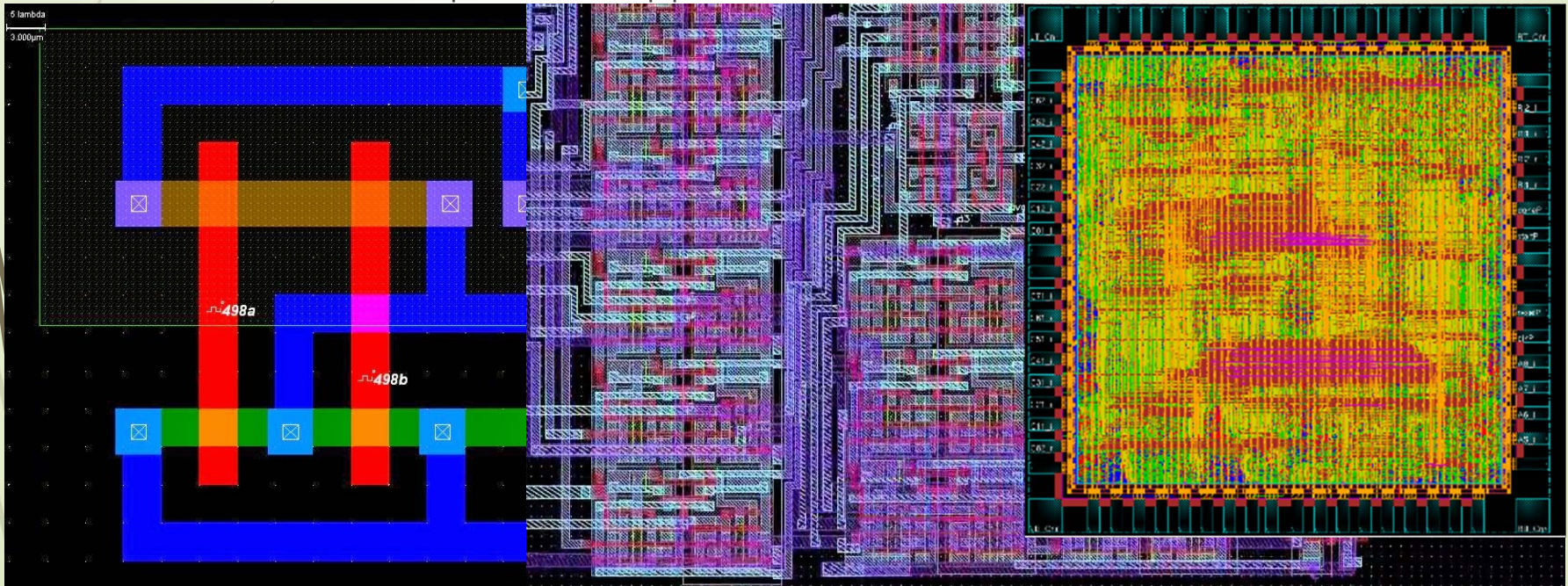
- Theoretical
- Abstract Level Definitions
- No Direct Connection with Silicon
- Suitable for Complex Circuits (Almost each design today)

Bottom – Up

- Starts at Silicon
- Builds Primitives
- Combined to get complex logic blocks
- Suitable for small projects

VLSI Chip Types

- Full Custom
 - Every circuit is custom designed
 - Time consuming, high initial cost
 - Suitable for mass-production
- Application Specific Integrated Circuits (ASICs)
 - ICs for a particular application



Design Metrics (1/2)

- The design quality is measured in terms of the following design metrics
 - Functionality
 - What operations it can perform on number of I/Os etc.
 - Cost (non-recurring and recurring)
 - NRE (Non-Recurrent Engineering) costs
 - One time cost factor like design time, design effort etc.
 - R&D, Infrastructure building, training, manufacturing equipment, VLSI CAD tools
 - Recurrent Costs
 - Proportional to volume, chip area
 - Silicon processing, packaging, testing
 - Reliability
 - noise margin/immunity

Design Metrics (2/2)

- Power dissipation (speed, power, energy)
 - Peak and average power dissipation,
 - Energy
- Speed (delay, operating frequency)
 - Delays between process input and output
 - Maximum operating frequency
- Time-to-market
 - An integrated circuit must be designed, verified, and finally implemented
 - As quick as possible to become available first in the market to beat the competitors.

Design Domains (Y-chart)

- The IC design can be described in the following three domains:

- Behavioral

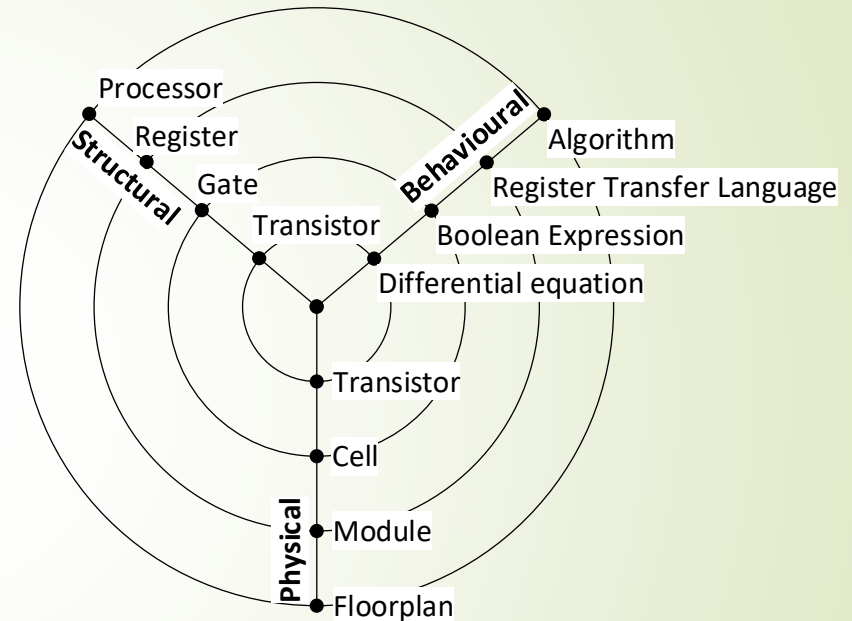
- Circuit is described by its behavior
- And not physical implementation or structure

- Structural

- Circuit is described by components and interconnections

- Physical

- Deals with actual geometry
- Described by shape, size and location of components



Gajski-Kuhn Y-chart

Computer Aided Design (CAD)

- Since the technology has allowed to reach today's integration levels
 - We need to use CAD tools to cope with the design complexity
 - A simple three loop circuit calculation requires three equations to be solved
 - Imagine the analysis/design complexity for millions and billions of transistors
- CAD tools fully or partly automate the VLSI design steps

Implementation Tools	Verification Tools
Logic and Physical Synthesis	Simulation
Design for Test (DFT)	Timing analysis
Full custom layout	Formal Verification
Floor planning	Power Analysis
Place and route	DRC and LVS

Preview of Lecture # 02

- ▶ Logic Design using MOSFETs
- ▶ MOSFETs as switches
- ▶ Logic Gates in CMOS